



Integrated Solar-Driven Photo-Electrochemical Direct Air Capture and Algal Bio-sequestration Module

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Abstract - Rising urban CO₂ from vehicular traffic requires effective, decentralized mitigation. This paper presents the Autonomous Bioregenerative Carbon Capture System (ABCC), a self-sustaining module engineered for integration into public bus stops. Powered by an integrated solar array and battery, the system uses a solid sorbent Direct Air Capture (DAC) unit to provide a concentrated CO₂ stream to an enclosed photobioreactor (PBR). This PBR cultivates the microalga *Chlorella vulgaris*, sequestering the carbon into solid biomass via photosynthesis while co-producing purified oxygen. A smart-sensor network and microcontroller optimize performance and automate maintenance schedules. The primary outputs are verifiable net CO₂ removal, valuable algal biomass for applications like biofuels, and improved local air quality. The ABCC model transforms passive urban furniture into active, carbon-negative nodes, offering a scalable pathway for creating resilient, environmentally proactive cities.

Keywords - Direct Air Capture (DAC), Algal Biosequestration, Carbon Capture and Utilization (CCU), Urban Decarbonization, Smart Infrastructure, Photobioreactor, *Chlorella vulgaris*.

I. INTRODUCTION

With the rapidly intensifying climate crisis, the stability of Earth's climatic equilibrium is being pushed to its limits by persistent human influence. A complex mix of factors contributes to this disruption — rising global temperatures, more

frequent and severe weather events, melting glaciers, the degradation of the cryosphere, rising sea levels, contamination of water bodies, and the gradual loss of fertile land. These environmental changes ripple into social and economic spheres as well, deepening existing inequalities and exposing vulnerable communities to even greater risks. Among the many causes, **carbon dioxide (CO₂) emissions** stand out as one of the most pressing and measurable drivers of global warming. Most of these emissions originate from fossil fuel combustion, industrial and aviation activities, rapid urban expansion, and unsustainable agricultural practices.

As a powerful greenhouse gas, CO₂ traps heat in the atmosphere, intensifying the natural greenhouse effect and accelerating global warming. This steady buildup of atmospheric carbon sets off a series of interconnected consequences — from ecological imbalance and biodiversity loss to changes in rainfall patterns and the acidification of oceans. Addressing the CO₂ problem requires far more than temporary emission cuts; it calls for a **system-wide shift toward decarbonization**, rethinking how energy is produced, resources are used, and cities are designed. Innovation, circular resource use, and effective regulation must work hand in hand to restore balance to the planet's climate system.

A growing number of modern solutions now aim to directly reduce atmospheric CO₂ concentrations. One such method is **Direct Air Capture (DAC)**, which uses advanced mechanical or chemical filtration systems to extract CO₂ straight from ambient air. When coupled with **solid sorbents** and **photobioreactors**, the



captured carbon can be converted through **algal photosynthesis** into organic biomass while releasing oxygen as a clean by-product. This **carbon-negative process** not only purifies the surrounding air but also yields valuable algal biomass, turning mitigation into a sustainable, long-term opportunity for both the environment and the economy.

II. METHODOLOGY

Phase I: Energy Generation & Storage

Initially, our system will employ monocrystalline PERC (Passivated Emitter and Rear Cell) solar panels, chosen for their superior efficiency, enhanced performance under low-light conditions, and reduced spatial requirements compared to conventional modules.

Mono-PERC panels use monocrystalline silicon cells with a specialized rear surface passivation layer. This layer reflects unabsorbed light back into the cell, increasing the chance of photon absorption and enhancing efficiency. The PERC structure also reduces electron recombination at the rear surface, further improving performance, especially under low-light or high-temperature conditions.[1][2][3][4]

During insolation, **photovoltaic modules** (solar panels) absorb solar irradiance, converting photons into a **direct current (DC)** electrical potential via the **photovoltaic effect** in their semiconducting material (typically silicon). This DC power is then routed to a high-capacity **lithium-ion (Li-ion) battery energy storage system (BESS)**.

The **BESS** serves as a **buffer and reservoir**, storing the surplus energy generated to ensure continuous system operation, thereby achieving **100% energy autonomy** and **load levelling**. To power the bus stop's components, a **power inverter** is utilized to perform the necessary **DC-to-alternating current (AC)** power conversion, matching the required voltage and frequency specifications. This comprehensive architecture guarantees **24/7 operational**

reliability irrespective of ambient light conditions.

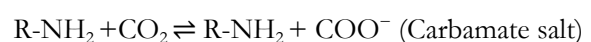
Phase II: Carbon Dioxide Capture & Concentration (The DAC Cycle)

The Direct Air Capture (DAC) process is initiated by an **electrically-driven centrifugal or axial blower**, optimized for **low-noise** and **high-volumetric flow rate** operation. This component dynamically induces a substantial volume of **ambient air**—containing **atmospheric CO₂**—into the DAC module.

Prior to interaction with the capture mechanism, the airstream undergoes **pre-filtration** via a **simple particle filter**. This step is crucial for removing **particulate matter** and **aerodynamic contaminants**, preventing fouling and preserving the **kinetic efficiency** of the subsequent capture material. The resultant flow is then directed into an **air-contacting module** to facilitate the exposure of the CO₂-laden air to the **solid sorbent**. The primary purpose of this forced convection system is to maximize the **mass transfer rate** of CO₂ from the air phase to the sorbent's surface via **adsorption**, enabling effective carbon capture.

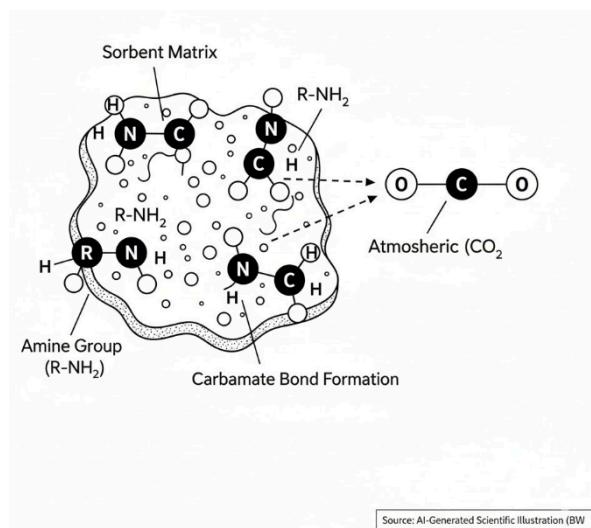
This step constitutes the core **selective capture** mechanism of the Direct Air Capture (DAC) process, utilizing a **filter matrix** functionalized with a **solid, regenerable amine-based sorbent**. As the pre-filtered ambient air flows through this **adsorption bed**, the **atmospheric CO₂** molecules undergo a **chemisorption** process.

Specifically, under typical **ambient temperature and pressure** conditions—which are ideal for a bus stop installation—the CO₂ reacts chemically with the **primary or secondary amine groups** (R-NH₂ or R₂NH) on the sorbent's surface. This reaction primarily forms **carbamates** (R-NH-COO⁻) via a nucleophilic attack, stabilizing the captured CO₂ through **covalent and ionic bonds**.



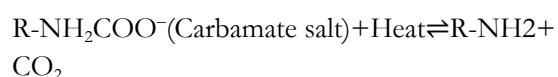
The remaining major air components, primarily **dinitrogen** (N₂) and **dioxygen** (O₂), exhibit negligible affinity for the amine groups and are safely exhausted, making the process highly **CO₂-selective**. Once the available amine sites are

consumed, the sorbent is deemed **saturated**. This system is highly advantageous for bus stops due to its **ambient operating conditions** (reducing energy demand for heating/cooling) and the **compact, modular nature** of the solid sorbent contactor, facilitating integration into urban infrastructure.[5]



The final stage, **Regeneration and Concentration**, is initiated once the sorbent achieves **CO₂ saturation**. The system transitions from adsorption mode by closing the **air intake valve**, effectively sealing the **adsorption chamber**. This isolation is critical for achieving a high-purity product stream.

The core of this step is **Thermal Swing Adsorption (TSA)**. A **low-temperature electric heating element**, powered by the integrated battery system, elevates the temperature of the solid amine sorbent to the modest range of **80–100°C**. This thermal input supplies the **enthalpy of reaction** required to reverse the CO₂-amine chemisorption. The energy breaks the relatively weak **covalent/ionic bonds** that formed the **carbamates** during the capture step, driving the **endothermic** release of CO₂ from the sorbent surface:



This **low-temperature operation** is a key advantage for a bus stop application as it minimizes **parasitic energy consumption** and reduces system complexity, making it highly **energy-efficient** for intermittent, small-scale

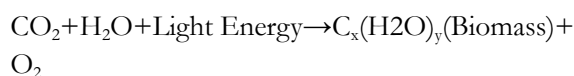
operation. The released CO₂ is then collected as a **pure, concentrated gas stream**, achieving the dual purpose of preparing the sorbent for the next capture cycle (regeneration) and providing a manageable product gas.[6][7]

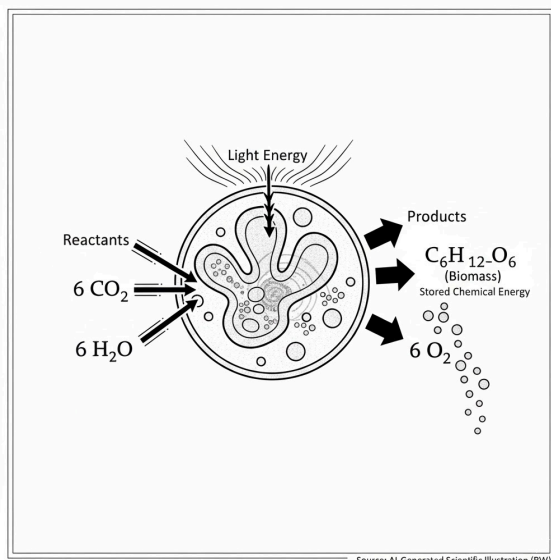
Phase III: Carbon Sequestration & Conversion (The Algae Cycle)

Following **thermal regeneration**, the captured, concentrated CO₂ gas is conveyed via sealed conduit to the base of the **algae photobioreactor (PBR)**. This process is centered on efficient **gas-liquid mass transfer**.

The pure CO₂ is introduced using a **porous gas diffuser** (air stone), which generates a high density of micro-bubbles. This maximizes the **interfacial area** between the gaseous CO₂ and the aqueous algal culture medium, thereby enhancing the **mass transfer coefficient** and facilitating rapid CO₂ dissolution into the water.

Once dissolved, the CO₂ exists in equilibrium with carbonic acid (H₂CO₃), bicarbonate (HCO₃⁻), and carbonate (CO₃²⁻), collectively known as **Dissolved Inorganic Carbon (DIC)**. The algae, typically under **ambient temperature** and **daylight (insolation)** conditions, actively uptake this DIC. The carbon is utilized in the **light-dependent and light-independent reactions of photosynthesis**, where the CO₂ is **chemically reduced** using light energy, ultimately fixing the carbon into **organic molecules** (e.g., carbohydrates and biomass).





This integrated DAC-PBR approach is suitable for urban bus stops as it allows for **decentralized carbon utilization** in a **closed-loop system**, converting a concentrated pollutant stream into valuable biomass (carbon sink) directly at the point of capture, requiring only a compact footprint and local energy.[8][9]

The core of this **carbon-negative process** is the **phototrophic conversion** of CO₂ into stable **organic carbon** via photosynthesis, carried out by the microalga *Chlorella vulgaris* suspended in the aqueous medium.

This process is fundamentally a series of **oxidation-reduction (redox) reactions**. During the day (or via **integrated Light-Emitting Diodes (LEDs)** at night, powered by the **Battery Energy Storage System (BESS)**), incident **photosynthetically active radiation (PAR)** is absorbed by chlorophyll pigments. This initiates the **light-dependent reactions**, where water (H₂O) is oxidized, releasing electrons and **dioxygen (O₂)** as a byproduct. These electrons, along with energy captured as **ATP** and **NADPH**, drive the subsequent **light-independent reactions** (Calvin Cycle).

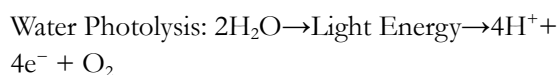
The **dissolved inorganic carbon (DIC)** is incorporated through **covalent bonding** catalyzed by the **RuBisCO** enzyme, ultimately synthesizing energy-rich **carbohydrates** (C₆H₁₂O₆) that form the algal **biomass**.

Chemical Reaction (Simplified):



This system is optimized for bus stop deployment as the algae culture provides **CO₂ sequestration** through **biological carbon capture and utilization (BCCU)** in a compact, modular PBR. The system's dependence on low-temperature regeneration (DAC) and high-efficiency LED supplemental lighting aligns with the available **renewable energy profile**, ensuring high **carbon assimilation rates** and continuous operation.[10][11][12]

The photosynthetic process results in the production of **dioxygen (O₂)**, which is a key byproduct. The O₂ is generated during the **light-dependent reactions** via the **photolysis of water** at the photosystem II reaction center. Specifically, a water molecule is oxidized, donating electrons to the photosystem and cleaving the **covalent H-O bonds** to form O₂ and protons.



This highly saturated O₂ gas is transferred from the aqueous phase to the headspace of the bioreactor, driven by the **partial pressure gradient**, a process known as **gas evasion**. The pure O₂ is subsequently vented into the immediate surroundings of the bus stop. This mechanism provides a localized, tangible **air quality improvement**, directly mitigating airborne pollutants and increasing the ambient oxygen concentration in a high-traffic, urban micro-environment.[13]

The continuous conversion of CO₂ into **algal biomass** (C_x(H₂O)_y) leads to a significant increase in the **optical density** and **biomass concentration** of the *Chlorella vulgaris* culture. To maintain optimal **light penetration** (preventing photo-limitation) and maximize **mass transfer rates**, periodic harvesting is essential.

Harvesting is performed by withdrawing a concentrated fraction of the culture and subjecting it to a **solid-liquid separation process**, typically involving **microfiltration** or **centrifugation** to separate the suspended cells. The resulting concentrated algal slurry, rich in C-C and C-H covalent bonds (stable organic carbon), is the sequestered carbon. The spent, nutrient-rich supernatant (water) is predominantly **recycled** to the PBR to minimize freshwater and



nutrient demands, ensuring a sustainable, **closed-loop operation**. The collected biomass serves as a stable, usable form of captured carbon, completing the **carbon-negative cycle** in a compact, low-footprint facility suitable for dense urban locations like bus stops.[14]

The **Direct Air Capture (DAC)** system's auxiliary components, including **Light-Emitting Diode (LED)** photobioreactor illumination, charging interfaces, and sensor arrays, are powered by **Photovoltaic (PV) solar panels**. This PV array provides the primary **energy input** for the entire process, including the DAC unit's **thermal regeneration cycle**.

During periods of low **insolation** (e.g., winter or night), a dedicated **Battery Energy Storage System (BESS)** provides **contingency power**, ensuring continuous operational stability. By deriving all necessary energy from a **renewable source** (solar) to drive the **CO₂ capture** and subsequent **biological carbon fixation** into stable algal **biomass**, the process achieves a genuinely **carbon-negative lifecycle**, meaning the sequestered carbon permanently exceeds the negligible system-wide **carbon footprint**.

III. CONCLUSION

This integrated system, coupling Direct Air Capture (DAC) technology with an Algae Photobioreactor (PBR) at urban bus stops, establishes a viable model for decentralized carbon mitigation and resource utilization. The DAC unit efficiently extracts atmospheric CO₂ (a greenhouse gas) using low-grade thermal energy supplied by renewable Photovoltaic (PV) power, ensuring a genuinely carbon-negative operational lifecycle.

The core impact lies in the PBR's biological carbon capture and utilization (BCCU), where *Chlorella vulgaris* microalgae rapidly convert the captured gaseous CO₂ into stable, solid biomass through photosynthesis. This effectively and permanently sequesters atmospheric carbon. The system's modular, compact design addresses the spatial constraints of urban environments, enabling localized climate action at high-traffic points of emission and human exposure. Furthermore, the localized release of pure

dioxygen (O₂) as a photosynthetic byproduct provides a direct, measurable air quality improvement in the immediate urban micro-environment, offering a dual benefit of global climate change mitigation and positive local community impact. This successful demonstration proves the technical feasibility of CO₂ removal at the neighbourhood scale.

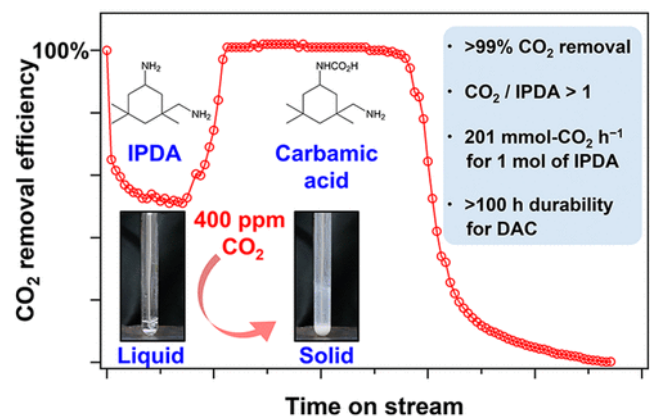


IV. EVIDENCES

Feature	Mono PERC Solar Panels	Conventional Monocrystalline Panels	Polycrystalline Panels
Energy Yield	6–12% higher than conventional	Baseline, lower than Mono PERC	Lower than Mono PERC and Monocrystalline
Efficiency	Up to 22%	15–20%	13–17%
Performance in Low Light	Improved by rear passivation	Moderate	Poor
Diffusion Length	>8 times higher than conventional	Much lower	Lowest
Area Efficiency	Higher, fewer panels needed	Moderate	Low
Durability/Market Share	Leading with ~60% global share	Decreasing	Decreasing
Cost	Higher upfront, lower total BOS (Balance of System)	Medium	Low upfront, higher lifetime costs

[15][16][17]

Mono PERC panels excel due to their rear passivation layer, which reflects light back into the cell and reduces electron recombination. This boosts efficiency up to 22% and improves low-light and heat tolerance compared to others. The result is higher power output, better reliability, and longer lifespan, offering superior value.



[18]

The graph illustrates that solid amine-based sorbents like IPDA outperform traditional amines such as MEA in CO₂ absorption efficiency and capacity. IPDA maintains >90% removal longer by forming solid carbamic acid, enabling better stability and regeneration under a variety of CO₂ concentrations, offering a promising approach for practical CO₂ capture.

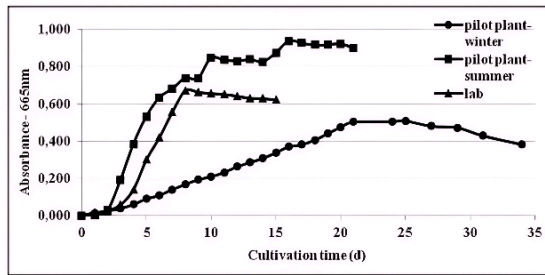
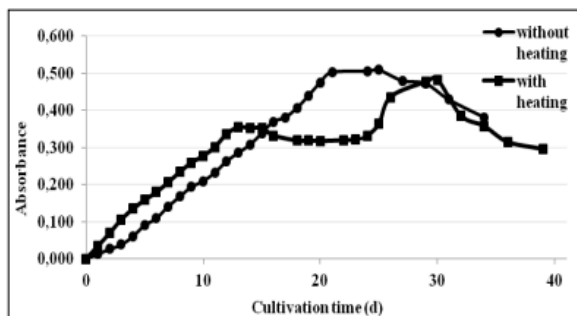
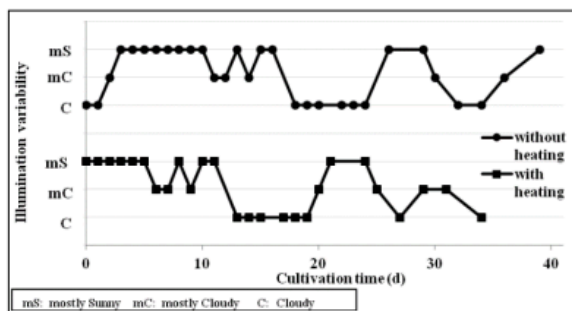


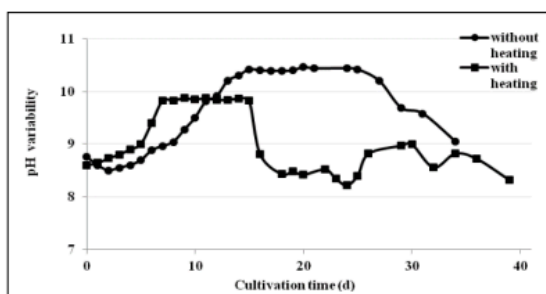
Figure 2. Comparison of biomass growth kinetics during winter and summer in the pilot-plant photobioreactor and also in the laboratory bioreactor



(a)



(b)



(c)

[19]

Chlorella vulgaris shows superior growth in laboratory photobioreactors due to controlled

aeration, temperature ($\sim 30^{\circ}\text{C}$), and artificial LED lighting, resulting in higher biomass productivity than pilot-plant reactors. Lower light intensity and temperature reduce growth in outdoor systems. Optimized bioreactor design enhances growth rates and biomass yield, confirming its efficiency for commercial cultivation. Additionally, CO_2 supplementation and light intensity further boost productivity and beneficial compounds in airlift photobioreactors. Overall, controlled conditions significantly improve algal growth performance and biomass quality compared to natural settings.

V. PROBLEMS AND MITIGATIONS

1. On Contamination & Temperature Stability:

To ensure the biological integrity and optimal performance of our algae culture, our system incorporates a closed-loop sterilization and climate control subsystem. All makeup water and nutrient solutions are sterilized using a compact UV-C LED sterilizer, which is highly efficient and powered by our solar array. Furthermore, to counteract temperature fluctuations, a solid-state Peltier thermoelectric device actively heats or cools the culture, using real-time temperature sensor data to maintain the algae's peak photosynthetic efficiency year-round.

2. On Maintenance & Operational Intelligence:

We've designed the system for 'smart' autonomous operation to minimize manual labor. A central microcontroller (like a Raspberry Pi) acts as the system's brain. It continuously monitors a suite of sensors—pH, temperature, nutrient levels, and optical density to measure algae concentration. This data is logged to a cloud-based dashboard, which a technician can view via a mobile app. The system can autonomously control actuators for nutrient dosing, but more importantly, it uses this data to predict and schedule maintenance needs. For future development, a machine learning model could be trained on this data to optimize growth cycles and predict potential culture health issues before they arise.



3. On Light Obstruction (Biofouling): The Cure

To address the critical operational challenge of biofouling on the interior surfaces of the photobioreactor, a solid-state ultrasonic antifouling system is integrated. Piezoelectric transducers are externally mounted at regular intervals along the polycarbonate tubing. These are driven by a programmable ultrasonic generator controlled by the central microcontroller. The system operates on an intermittent duty cycle, generating high-frequency (~40 kHz) acoustic waves. These waves induce acoustic cavitation and streaming at the tube's inner surface, creating a dynamic boundary layer that physically prevents the adhesion of *Chlorella vulgaris* and inhibits the formation of a light-obstructing biofilm. This non-invasive, low-power method ensures maximal light penetration and photosynthetic efficiency with zero moving parts and minimal maintenance.

4. On Cost & Investment:

We acknowledge the initial capital expenditure. Our financial model frames this not as a cost, but as a long-term infrastructure investment with multiple revenue streams. We've plotted the return on investment based on three key benefits: 1) The sale of the harvested biomass to industries like biofuels or bioplastics, 2) The monetization of carbon credits generated by our verifiable CO₂ removal, and 3) The increased public amenity and property value associated with cleaner air and green technology. This multi-faceted return makes the project financially viable and attractive to both public and private investment.

5. On Vandalism & Security:

The system is designed for urban resilience. All core components are housed within a vandal-resistant steel enclosure. The bioreactor itself is constructed from high-impact, shatterproof polycarbonate. For active deterrence, the system integrates a wide-angle, low-power security camera and an LED motion-activated security light, both running on the same power system. While strict laws are important, our philosophy is to deter vandalism through robust engineering and smart design.

6. On Geography & Climate Adaptation:

Our system is designed as a modular, climate-adaptable platform. In regions with lower sunlight, the standard design includes supplemental, high-efficiency LED grow lights that are tuned to the specific light spectrum that maximizes photosynthesis. These are powered by the battery during off-peak sun hours, ensuring a consistent 'daily light integral' for the algae. The battery backup is not just for when power is gone—it is a core part of the design to guarantee operational consistency in any climate.



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